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Semiconductor device including a memory hub with chiplet structure

Abstract

Disclosed is a semiconductor device including a memory hub with a chiplet structure. The semiconductor device includes an interposer, a processor unit placed on the interposer, at least one memory hub placed on the interposer and provided to be physically separated from the processor unit, and at least one memory provided to be physically separated from the memory hub.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to and the benefit of Korean Patent Application No. 10-2024-0112570, filed on Aug. 22, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

1. Technical Field
- (2) The present disclosure relates to a semiconductor device including a memory hub with a chiplet structure, and a semiconductor design structure. More particularly, the present disclosure relates to a semiconductor design technology for a chiplet-based memory hub using a die-to-die (D2D) technology, which can maximize effective memory bandwidth during on-device artificial intelligence (AI) computation.
2. Related Art
- (3) Integrating a large number of functions into a single chip requires a large chip area, thus physically increasing the number of wires needed, thereby making bandwidth expansion difficult. In addition, unnecessary power consumption increases, thus increasing heat generation, requiring additional design and cost for efficient thermal management. Furthermore, due to the integration of various functions, power management and optimization become complicated, which may cause

inefficient operation.

(4) In addition, fabricating a semiconductor including a memory and an interface as a single chip significantly increases manufacturing cost when using the latest process node. As the area of a single chip increases, the probability of defect occurrence during the fabrication process increases, resulting in a higher defect rate and lower yield, thereby increasing the unit manufacturing cost. Moreover, an increased number of process steps required for fabricating a complex single chip results in an increase in overall manufacturing cost.

(5) Furthermore, in a memory interface semiconductor manufactured through a conventional bumping process, bumps occupy a relatively large area. Particularly, as a data path extending out of the chip through a physical conductor such as a printed circuit board (PCB) increases, bandwidth per unit area is limited, and the increased length of the data transmission path causes a reduction in transmission speed, resulting in degradation of the system performance.

(6) With the emergence of consumer memories operating at ultra-high frequencies, there is a need for a new chiplet-based memory hub structure and semiconductor design technology capable of enabling high-speed data transmission between chips and realizing data transmission with high bandwidth and low latency, in order to prevent memory bottlenecks occurring in various on-device computation-intensive applications.

SUMMARY

(7) Embodiments of the present disclosure are directed to providing a semiconductor device including a memory hub with a chiplet structure, and a semiconductor design structure.

(8) Furthermore, embodiments of the present disclosure are directed to providing a semiconductor device and a design technology, in which a memory hub is configured based on a chiplet structure, which can connect and form a plurality of individual dies, thereby enabling each chiplet to be fabricated at an optimal process node to achieve defect rate reduction and cost reduction, allowing continuous performance improvement through upgrade and replacement of individual chiplets, and reducing latency through direct connection between chips using a die-to-die (D2D) interface.

(9) Furthermore, embodiments of the present disclosure are directed to providing a semiconductor device and a design technology, which enable direct connection between chips through a D2D interface configured with small-sized micro bumps (uBumps) on an interposer, thereby reducing space required per unit area and enabling high-bandwidth connection between chiplets, realizing data transmission with high low bandwidth and latency, improving efficiency and performance by integrating individual chiplets through the interposer, and achieving reduced power loss and improved power efficiency due to shorter physical distances compared to conventional structures.

(10) In addition, embodiments of the present disclosure are directed to providing a memory interface and memory hub configuration, which adopts a structure such that micro bump (uBump)-based consumer memories are placed on the interposer, making it possible to increase routing density through efficient spatial arrangement, thereby reducing physical distances and thus minimizing signal distortion and latency, enhancing performance and reducing upgrade and development costs through flexible support for increasingly advanced consumer memories operating at ultra-high frequencies, satisfying individual requirements by using consumer memories suited for various applications, and preventing memory bottlenecks occurring in on-device computation-intensive applications through support for high-frequency memories.

(11) The problems to be solved by the present disclosure are not limited to those mentioned above, and other technical problems not specifically mentioned will be clearly understood by those skilled in the art from the following description.

(12) A semiconductor device including a memory hub with a chiplet structure according to embodiments of the present disclosure may include: an interposer; a processor unit placed on the interposer; at least one memory hub placed on the interposed and provided to be physically separated from the processor unit; and at least one memory placed to be physically separated from the at least one memory hub.

- (13) Furthermore, the processor unit may include a processor and a die-to-die (D2D) interface unit. The at least one memory hub may include a D2D interface unit and a memory controller. The processor unit and the at least one memory hub may be connected through the D2D interface units.
- (14) Furthermore, the D2D interface unit may include a D2D physical interface (PHY) provided at a lower portion, and a D2D controller provided at an upper portion.
- (15) In addition, the D2D PHY may include a plurality of micro bumps placed on the interposer, and each of the plurality of micro bumps may have a diameter ranging from 30 μm to 60 μm .
- (16) Also, the at least one memory may be packaged with a plurality of micro bumps. The at least one memory may be connected to the at least one memory hub through the interposer.
- (17) In addition, the at least one memory may not be packaged with the micro bumps. The semiconductor device may further include a sub-interposer provided between the interposer and the memory. A plurality of micro bumps may be provided under the sub-interposer.
- (18) Furthermore, the processor unit and the at least one memory hub may be connected through wires arranged in the interposer, and the at least one memory hub and the at least one memory may be connected through wires arranged in the interposer.
- (19) Furthermore, the processor unit and a plurality of memory hubs may be connected, and each of the plurality of memory hubs may be connected to a plurality of memories.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a conceptual diagram for explaining a structure of a semiconductor device and a semiconductor package according to a conventional memory interface structure.
- (2) FIGS. 2A and 2B are a perspective view and a cross-sectional view illustrating a semiconductor device that forms a memory hub with a chiplet structure according to an embodiment of the present disclosure.
- (3) FIG. 3 is an exemplary view for explaining a configuration of a semiconductor device that forms a memory hub with a chiplet structure according to another embodiment of the present disclosure.
- (4) FIG. 4 is an exemplary view comparatively illustrating the size of a micro bump (uBump) according to an embodiment of the present disclosure.
- (5) FIG. 5 is a conceptual diagram for explaining a configuration of a memory hub system with a chiplet structure according to an embodiment of the present disclosure.
- (6) FIG. 6 is a flowchart for explaining a memory operation method in the memory hub system according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

- (7) Hereinafter, the present disclosure will be described in detail with reference to the accompanying drawings so that those skilled in the art can easily practice the present disclosure. However, the present disclosure may be embodied in various different forms and is not limited to the embodiments described herein.
- (8) The terms used herein are intended to describe the embodiments and are not intended to limit the present disclosure. In this specification, a singular form also includes a plural form unless specifically stated otherwise in a phrase.
- (9) As used herein, the terms “comprise” and “comprising” are intended to denote the existence of components, steps, operations, and/or elements described in the specification, and do not exclude the presence or addition of one or more other components, steps, operations, and/or elements.
- (10) In the following description of the present disclosure, a detailed description on well-known arts which is determined to make the gist of the present disclosure unclear will be omitted.
- (11) Hereinafter, embodiments according to the present disclosure will be described in detail with

reference to the attached drawings. The configuration of the present disclosure and operational effects thereof will be clearly understood through the following detailed description.

(12) FIG. 1 is a conceptual diagram for explaining a structure of a semiconductor device and a semiconductor package according to a conventional memory interface structure.

(13) Referring to FIG. 1, according to the conventional memory interface structure, a single chip **10** formed of one die includes a processor **10a** including, for example, a central processing unit (CPU), a graphics processing unit (GPU), a neural processing unit (NPU), or the like, a memory controller **10b**, and a memory physical interface (PHY) **10c**. The single chip **10** may be connected to a memory **13** through wiring connections via a lower package substrate **15** through a plurality of bumps **12** disposed on an interposer **14**, and further via a plurality of package bumps **11** disposed on a printed circuit board (PCB) **16**.

(14) In such a case, since a processor function and a memory control function are integrated into the single chip **10**, a required area is increased, thus physically increasing the number of wires needed, thereby making bandwidth expansion difficult. Additionally, unnecessary power consumption increases, thus increasing heat generation, requiring additional design and cost for efficient thermal management. Furthermore, due to the integration of various functions, power management and optimization become complicated, which may cause inefficient operation.

(15) In addition, in an existing memory interface semiconductor, the package bumps **11** such as solder ball bumps occupy a large area within the PCB **16** connected to the memory. Particularly, a data path extending out of a chip through a physical conductor such as the PCB **16** becomes longer, thereby limiting bandwidth per unit area. Furthermore, the increased length of the data transmission path reduces transmission speed, which may result in degradation of system performance.

(16) FIGS. 2A and 2B are a perspective view and a cross-sectional view illustrating a semiconductor device that forms a memory hub with a chiplet structure according to an embodiment of the present disclosure.

(17) Referring to FIGS. 2A and 2B, a memory hub system is configured through a chiplet structure in which a processor unit **100** and a memory hub **200** are formed as individual dies and electrically connected to each other. As the chiplet-based individual dies are fabricated at respective optimal process nodes, a defect rate can be reduced, and manufacturing cost can be lowered. Furthermore, continuous performance improvement is achievable through upgrades and replacements of the individual chiplet structures, and latency can be reduced through a direct chip connection through a die-to-die (D2D) interface.

(18) Referring to FIGS. 2A and 2B, a semiconductor device including a memory hub with a chiplet structure may include an interposer **140**, a processor unit **100** placed on the interposer **140**, one or more memory hubs **200** placed on the interposer **140** and provided to be physically separated from the processor unit **100**, and one or more memories **300** provided to be physically separated from the memory hubs **200**. The interposer **140** may be placed on a package substrate **150**. Furthermore, the processor unit **100**, the memory hubs **200**, and the memories **300** may be arranged on the interposer **140** in a sideways arrangement rather than being stacked, so that the individual chips can be integrated on the single interposer **140**, thereby improving efficiency and performance.

(19) Here, the processor unit **100** may include a processor **110** configured as a CPU, a GPU, an NPU, or the like, and a D2D interface unit **120** for connection with the memory hubs **200**, which are physically separated and formed as individual dies. Each of the memory hubs **200** may include a D2D interface unit **210** for connection with the processor **110** formed as an individual die, and a memory controller **220** configured to control the connected memory. The processor unit **100** and the memory hub **200** may be connected to transmit and receive data through the respective D2D interface units **120** and **210**. The memory controller **220** of the memory hub **200** may include a memory controller **221** provided at an upper portion and a memory physical interface (PHY) **222** provided at a lower portion.

(20) Furthermore, the D2D interface unit **120** of the processor unit **100** may include a D2D

physical interface (PHY) **122** at a lower portion, and a D2D controller **121** at an upper portion. Similarly, the D2D interface unit **210** of the memory hub **200** may include a D2D PHY **212** at a lower portion, and a D2D controller **211** at an upper portion.

(21) In addition, the D2D PHY **122** and **212** may include a plurality of micro bumps **130** placed on the interposer **140**. For example, each of the micro bumps **130** may have a diameter ranging from 30 μm to 60 μm . The memory **300** connected to the memory hub **200** may be packaged with a plurality of micro bumps. The memory **300** may be connected to the memory hub **200** through the memory PHY **222** via the interposer **140**. The processor unit **100** and the one or more memory hubs **200** may be connected through wires arranged in the interposer **140** via the micro bumps **130**. The one or more memory hubs **200** and the one or more memories **300** may also be connected through wires arranged in the interposer **140** via the micro bumps **130**.

(22) As described above, the processor unit **100** and the memory hubs **200**, which are formed as individual dies, may be directly connected through the D2D interface configured with the micro bumps on the interposer **140**, thus reducing required space per unit area, thereby allowing data transmission with high bandwidth and low latency between chips. Furthermore, the individual chips may be integrated through the interposer, so that efficiency and performance may be improved, and compared to the conventional substrate structure, power loss may be reduced by a reduced physical distance, thereby enhancing power efficiency. In addition, micro bump-based consumer memories may be placed on the interposer to increase routing density through efficient spatial arrangement, thereby reducing physical distances and minimizing signal distortion and latency. Increasingly advanced consumer memories operating at ultra-high frequencies may be flexibly supported, thereby reducing performance upgrade costs and satisfying the requirements of various applications. Here, the consumer memories each refer to an individual memory included in a memory in which a plurality of memory chipsets are used in a module form.

(23) For example, in the case of a memory **310** packaged with general bumps instead of micro bumps, a sub-interposer **160** may be additionally placed between the interposer **140** and the memory **310**, and a plurality of micro bumps may be provided under the sub-interposer **160**, thereby supporting memories having various bump specifications.

(24) Furthermore, the memories **300** and **310** may be implemented in various forms of memory, and for instance, may include a double data rate synchronous dynamic random access memory (DDR SDRAM), a low power double data rate (LPDDR) SDRAM, a graphics double data rate (GDDR) SDRAM, and a Rambus dynamic random access memory (RDRAM). The memories **300** and **310** may also be implemented as various forms such as a static random access memory (SRAM), a high bandwidth memory (HBM), or a processor-in-memory (PIM), and are not limited thereto.

(25) FIG. 3 is an exemplary view for explaining a configuration of a semiconductor device that forms a memory hub with a chiplet structure according to another embodiment of the present disclosure.

(26) Referring to FIG. 3, a single processor unit **100** may be connected to a plurality of memory hubs **201** and **202**. Each of the memory hubs **201** and **202** may be connected to a plurality of memories **301**, **302**, **310**, and the like. As such, according to the memory hub system structure of the present disclosure, a design structure capable of more freely expanding a plurality of memory hubs and memories using a D2D interface may be provided through flexible scalability.

(27) FIG. 4 is an exemplary view comparatively illustrating the size of a micro bump (uBump) according to an embodiment of the present disclosure.

(28) Referring to FIG. 4, each of the micro bumps **130** used in the present disclosure has a diameter ranging from 30 μm to 60 μm . Solder balls used in conventional ball grid array (BGA) type memories have a size of 400 μm or more, standard flip chip bumps used in flip chips have a size of 150 μm to 200 μm , and flip-pitch bumps have a size of approximately 100 μm .

(29) As described above, in the present disclosure, the consumer memory configured with the

micro bumps (uBumps) may be connected through the interposer, enabling routing to a greater number of memories per unit area compared to conventional ball-type memories, thereby enabling configuration of memory interface channels the number of which is n times or more the number of memory interface channels of the conventional ball-type memory. In addition, a higher routing density may be achieved compared to an interface configuration using the ball-type consumer memories in the same area. As a result, the physical distance between the consumer memory and the PHY can be reduced, and timing and skew compared to an advantage in matching conventional methods may be obtained. Furthermore, a power mesh may be more precisely configured in chipsets connected through the micro bumps, stable signal output.

(30) FIG. 5 is a conceptual diagram for explaining a configuration of a memory hub system with a chiplet structure according to an embodiment of the present disclosure.

(31) Referring to FIG. 5, a D2D interface-based memory hub system according to an embodiment of the present disclosure may include a processor unit **100**, a plurality of memory hubs **201** and **202**, and a plurality of memories **301**, **302**, **303**, and **304**.

(32) The processor unit **100** may include a processor **110** configured as a CPU, a GPU, an NPU, or the like, and a D2D interface unit **120** for a D2D interface. Each of the memory hubs **201** may include a D2D interface unit **210** for a D2D interface, a memory controller **221**, and a memory PHY **222**.

(33) The processor unit **100** may sense and recognize the plurality of memory hubs **201** and **202** and may recognize a status of a memory resource connected to each memory hub, for example, whether memory allocation is available (**S510**). The processor unit **100** may set a data path based on the status information of the memory resource recognized through each of the memory hubs **201** and **202** (**S520**). For example, based on the status information of the memory resource of the first memory hub **201** and the second memory hub **202**, it may be determined which one of the first memory hub **201** or the second memory hub **202** is to be used to perform data transmission.

(34) Memory allocation using the determined memory hub may be performed using a plurality of memories connected to each memory hub, so that data may be dynamically distributed (**S530**). Here, based on the resource status information of each of the plurality of memories connected to the corresponding memory hub, it may be determined which memory is to be used to perform data transmission. Data transmission may be performed using the determined memory (**S540**).

(35) FIG. 6 is a flowchart for explaining a memory operation method in the memory hub system according to an embodiment of the present disclosure.

(36) Referring to FIG. 6, the processor unit **100**, which is a main chipset, may detect a memory resource request (**S610**).

(37) Next, the processor unit **100** may analyze the memory request (**S620**), and may check memory usage conditions of the plurality of memory hubs **200** (**S630**).

(38) The processor unit **100** may determine an available memory (**S640**), and may allocate the corresponding memory resource (**S650**).

(39) The memory hub corresponding to the related memory resource may check memory status information (**S660**). The memory hub may allocate the memory (**S670**).

(40) Based on the allocated memory, a memory mapping and routing table may be updated (**S680**).

(41) It may be determined whether data has been normally received and whether the memory operation is normal (**S690**). If the operation is normal, the memory request processing may be completed (**S700**).

(42) According to the present disclosure, through an on-device AI consumer memory hub chiplet interface technology, a main processor such as a CPU, GPU, or NPU, which performs on-device AI computations that are expected to increase, may access a high-performance consumer memory via a D2D interface, thereby providing the effect of accelerating high-speed computation.

(43) Furthermore, according to the present disclosure, the combination of interposer-based next-generation consumer memory interface technology and D2D technology enables the

implementation of an ultra-high-speed and low-power data transmission technology for on-device AI computation data.

(44) In addition, according to the present disclosure, it is possible to implement a memory access technology through the D2D interface, which is essential for a chiplet configuration.

(45) The effects of the present disclosure are not limited to those described above, and other effects not mentioned will be clearly understood by those skilled in the art from the following description.

(46) The embodiments disclosed in the specification of the present disclosure are merely illustrative, and the present disclosure is not limited thereto. The scope of the present disclosure should be interpreted based on the following claims, and all technical ideas that fall within the range equivalent to the claims should be understood as belonging to the scope of the present disclosure.

Claims

1. A semiconductor device including a memory hub with a chiplet structure, the semiconductor device comprising: an interposer; a processor unit placed on the interposer; at least one memory hub placed on the interposer and provided to be physically separated from the processor unit; and at least one memory placed to be physically separated from the at least one memory hub, wherein the processor unit comprises a processor and a die-to-die (D2D) interface unit, the at least one memory hub comprises a D2D interface unit and a memory controller, and the processor unit and the at least one memory hub are connected through the D2D interface units, wherein the D2D interface unit of each of the processor unit and the at least one memory hub comprises a D2D physical interface (PHY) provided at a lower portion, and a D2D controller provided at an upper portion, wherein the D2D PHY includes a plurality of micro bumps placed on the interposer, and each of the plurality of micro bumps has a diameter ranging from 30 μm to 60 μm , wherein the at least one memory is packaged with a plurality of micro bumps having a same diameter as the plurality of micro bumps, and the at least one memory is connected to the at least one memory hub through the interposer, and wherein the processor unit and the at least one memory hub are directly connected through the micro bumps via wires arranged in the interposer, and the at least one memory hub and the at least one memory are connected through the micro bumps via wires arranged in the interposer.
 2. The semiconductor device as claimed in claim 1, wherein the at least one memory is not packaged with the micro bumps, the semiconductor device further comprising a sub-interposer provided between the interposer and the memory, wherein a plurality of micro bumps are provided under the sub-interposer.
 3. The semiconductor device as claimed in claim 1, wherein the processor unit and a plurality of memory hubs are connected, and each of the plurality of memory hubs is connected to a plurality of memories.
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